SECURITY OPERATIONS INCIDENT INVESTIGATION REPORT

Scenario 3 | Registry Run-key modification | Evidence-based review

Assessment	Evidence set	Event date / time standard
Confirmed Run-key modification Overall confidence: HIGH	23 archived records; 3 linked Wazuh alerts One Windows endpoint	27 October 2026 Event timestamps reported in UTC

Method: Conclusions below derive only from the two supplied JSONL files. Record notation **A#** and **L#** identifies the 1-based line number in the Archives and Alerts files respectively. Sysmon/Windows event record IDs are included for source verification.

1. Executive Summary

At **12:05:07.993 UTC on 27 October 2026**, Sysmon recorded a registry **SetValue** operation by **reg.exe** against a per-user Windows startup Run key, setting the value **Atomic Red Team** to **C:\Path\AtomicRedTeam.exe**. This is direct evidence of configuring a logon-triggered execution entry. Earlier process-creation records show the same reg.exe instance was started by cmd.exe, which was started by PowerShell; the account recorded for these events is **DESKTOP-5OG7C28\Darragh Casey** [A21-A23; L1-L3].

In the approximately 1.4 seconds preceding the write, the same PowerShell parent created four **hostname.exe** and two **whoami.exe** processes. Their parent process GUID, user, logon ID and timing support treating the discovery and Run-key change as one local command sequence [A15-A23].

The key modification and process relationships are **observed facts**. An intention to establish persistence is a **strong inference** from the autostart location and assigned executable path. The supplied records do **not** show that the executable exists, ran, survived a new logon or caused any subsequent compromise. The string "Atomic Red Team" and the isolated-lab context suggest test activity, but neither should be treated as proof of benign authorization.

2. Scope of Activity

Two supplied newline-delimited JSON sources were examined: **Scenario_3_Archives.json** (23 records) and **Scenario_3_Alerts.json** (3 alerts). All three alerts mirror Sysmon records already present in the archive, as shown by matching Windows channel, event record IDs and event contents: 28371, 28372 and 28373. Accordingly, the combined material contains **23 distinct underlying Windows events, not 26** [A21-A23; L1-L3].

Source/channel	Archive records	Types represented
Sysmon/Operational	11	10 x Event ID 1 (Process Create); 1 x Event ID 13 (Registry value set)
Windows Security	10	2 x 4624; 2 x 4672; 6 x 5379
Windows System	2	2 x Kernel-Power 566
Wazuh alerts (additional feed)	3 (duplicates)	Rule IDs 92052, 92041, 92302; duplicate Sysmon 28371-28373

All records are reported by Wazuh agent **001 / Windows_VM / 192.168.232.130** for Windows host **DESKTOP-5OG7C28**; Wazuh manager: **dcasey-virtual-machine**. The earliest underlying event is **12:02:05.825 UTC** and the latest is **12:05:07.995 UTC** (systemTime). The main related activity is concentrated at **12:05:06.591-12:05:07.993 UTC**. Sysmon eventdata.utcTime is used for the precise process/registry timeline below; Windows systemTime is used elsewhere. Wazuh ingestion timestamps carry a **+0100** offset and should not be confused with event occurrence time.

Visibility is limited to this one host and these supplied event types and time interval. No conclusions can be drawn about the broader environment, preceding PowerShell launch, earlier persistence state, later re-logon, payload

execution, external access or network communications from the absence of corresponding records in this subset.

3. Detailed Investigation

3.1 Local host and user discovery

Observed: Four hostname.exe process creations at 12:05:06.591, .650, .715 and 12:05:07.471 UTC (PIDs 2772, 11160, 1148, 8600), and two whoami.exe creations at 12:05:06.747 and 12:05:07.500 (PIDs 8076, 8176). Each has PowerShell PID **12276** and identical parent process GUID **{4b2ce92d-8a8c-6ae0-6206-000000001a00}**; each lists user DESKTOP-5OG7C28\Darragh Casey, logon ID **0xfcd15** and **High** integrity [A15-A20; Sysmon 28365-28370].

Inference: hostname.exe is consistent with obtaining the computer name, and whoami.exe is consistent with identifying the execution identity. Repetition in a short interval is consistent with scripted or manually repeated host/user discovery. There is no command-output capture or script body, so the outputs, operator intention and use of those outputs are unknown.

3.2 PowerShell-to-command-shell-to-registry chain

Observed: At 12:05:07.886 UTC, cmd.exe PID **7912** was created with the same PowerShell parent GUID, user, logon ID and High integrity. Its command line invokes **REG ADD** through **cmd.exe /c**. At 12:05:07.953 UTC, reg.exe PID **2824** was created with parent PID 7912 and parent GUID matching the cmd.exe process GUID. At 12:05:07.993 UTC, Sysmon Event ID 13 attributes the SetValue to that identical reg.exe PID **2824** and process GUID [A21-A23; L1-L3].

Process lineage (recorded GUID links):

```
powershell.exe  PID 12276  GUID {4b2ce92d-8a8c-6ae0-6206-000000001a00}
 |-- hostname.exe x4; whoami.exe x2       [A15-A20]
 `-- cmd.exe PID 7912                    [A21 / L1]
      GUID {4b2ce92d-9373-6ae0-c606-000000001a00}
      `-- reg.exe PID 2824               [A22 / L2]
           GUID {4b2ce92d-9373-6ae0-c806-000000001a00}
           `-- registry SetValue         [A23 / L3]
```

The process creation of PowerShell PID 12276 is **not** present in the supplied records. It is established here only as the repeated parent, not as proof of how PowerShell was launched or which script was run.

3.3 Registry modification and persistence significance

Observed: cmd.exe and reg.exe both include the following substantive registry arguments; Sysmon 13 independently confirms the exact registry target and stored data [A21-A23]:

```
REG ADD "HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run"
  /V "Atomic Red Team" /t REG_SZ /F /D "C:\Path\AtomicRedTeam.exe"
```

```
Sysmon Event 13: EventType=SetValue
TargetObject=HKU\S-1-5-21-2586188580-1629295405-1270050116-1001\Software\Microsoft\Windows\Curren
tVersion\Run\Atomic Red Team
Details=C:\Path\AtomicRedTeam.exe
Image=C:\WINDOWS\system32\reg.exe | PID=2824
```

The command uses the **HKCU** alias; the resulting event shows the corresponding user-hive location under **HKU\S-1-5-21-...-1001**. This agreement supports that the process acted on the user-level Run key. The **/F** option requests forced writing without prompting, but the telemetry does not state whether an old value was present or overwritten.

Inference: A value under CurrentVersion\Run directs Windows to launch the specified command at the user's future sign-in, subject to normal operating conditions. This demonstrates establishment of a **persistence mechanism/configuration**, not execution of the referenced executable. The value name and executable path are plainly visible and test-themed; the path looks illustrative, but there is no file existence or execution evidence with which to classify the target as real, valid, malicious or benign.

3.4 Wazuh alert validation and discarded assumptions

Alert / source	Assessment of underlying telemetry
92052 / Level 4 "Windows command prompt started by an abnormal process" [L1 / A21; Sysmon 28371]	Fact: cmd.exe was started by powershell.exe and carried the Run-key command. Calling that parent "abnormal" is a rule characterization, not independent proof of compromise. The embedded command line is the relevant evidence.
92041 / Level 10 "Value added to registry key has Base64-like pattern" [L2 / A22; Sysmon 28372]	Fact: this is a **process-create** event for reg.exe, not a registry-write record. Its command specifies a cleartext path, not a visible encoded payload. The title alone does not substantiate Base64 encoding, obfuscation (T1027) or a completed registry modification.
92302 / Level 6 "Registry entry to be executed on next logon was modified using ... reg.exe" [L3 / A23; Sysmon 28373]	Substantiated: Sysmon 13 directly records **SetValue**, the Run-key path, exact data and the reg.exe process GUID. It shows a configuration change, but does not prove a later logon or launch.

3.5 Other records and relevance decision

The six Security 5379 records (12:02:05.949-12:02:06.024 UTC) concern Credential Manager reads using client PID **7472**, with WindowsLive target names; four report zero returned and two report one returned [A3-A8]. They occur roughly three minutes before the Run-key change, and no supplied process event links PID 7472 to PowerShell PID 12276 or the registry-writing process. The account/context also includes logon IDs **0x3e7** and **0xfcd57**, unlike the core sequence's **0xfcd15**. They are retained as background, **not attributed to credential theft or the Run-key sequence**.

Security 4624/4672 records are SYSTEM/service-logon and privilege notifications (logon ID **0x3e7**) [A1-A2, A13-A14]. Sysmon shows MoUsoCoreWorker.exe and service-host activity under SYSTEM [A9, A12], and Kernel-Power 566 describes unlock/input session transitions [A10-A11]. None has a demonstrated process GUID, logon-ID or command-line causal connection to the registry-writing chain. These records do not establish privilege escalation. The core child processes having **High** integrity describes their execution context, not how that integrity was obtained.

4. Evidence Analysis

4.1 Primary evidentiary anchors

Evidence / source	Verifiable values	Contribution
E1: Repeated parent A15-A21; Sysmon 28365-28371	ParentImage: ...\powershell.exe; ParentPID: 12276; ParentGUID: {4b2ce92d-8a8c-6ae0-6206-000000001a00}; User: ...\Darragh Casey; LogonId: 0xfcd15	Links six discovery processes and cmd.exe to one known parent context.
E2: Command shell A21 = L1; Sysmon 28371	PID: 7912; GUID: {4b2ce92d-9373-6ae0-c606-000000001a00}; /c REG ADD HKCU\...\Run	Shows requested Run-key addition; direct command evidence.
E3: Registry tool A22 = L2; Sysmon 28372	PID: 2824; GUID: {4b2ce92d-9373-6ae0-c806-000000001a00}; ParentPID: 7912; ParentGUID equals E2 GUID	Proves cmd.exe-to-reg.exe parent-child linkage.
E4: Registry state change A23 = L3; Sysmon 28373	EventType: SetValue; PID: 2824; GUID equals E3 GUID; TargetObject: HKU\...\CurrentVersion\Run\Atomic Red Team; Details: C:\Path\AtomicRedTeam.exe	Strongest confirmation of actual value write and of attribution to E3 process.

Evidence granularity: Event ID 1 records show launches and command lines, not exit codes; Event ID 13 shows a registry value being set, not whether a later startup succeeded. Matching **process GUIDs** provides stronger linkage than PID reuse alone. E1-E4 refer to one local process branch; there is no evidence for remote command origin or a second endpoint.

4.2 Source traceability and duplication

Line	UTC time	Channel / ID	Record ID	Short description
A1	12:02:05.825	SEC / 4624	83539	Service logon (type 5)
A2	12:02:05.825	SEC / 4672	83540	Special privileges / SYSTEM
A3	12:02:05.948	SEC / 5379	83541	Credential Manager read
A4	12:02:05.949	SEC / 5379	83542	Credential Manager read
A5	12:02:05.949	SEC / 5379	83543	Credential Manager read
A6	12:02:06.023	SEC / 5379	83545	Credential Manager read
A7	12:02:06.023	SEC / 5379	83544	Credential Manager read
A8	12:02:06.023	SEC / 5379	83546	Credential Manager read
A9	12:02:53.350	SMN / 1	28363	MoUsoCoreWorker.exe
A10	12:03:53.553	SYS / 566	4086	Power session transition
A11	12:04:50.329	SYS / 566	4087	Power session transition
A12	12:04:50.376	SMN / 1	28364	svchost.exe
A13	12:04:50.375	SEC / 4624	83547	Service logon (type 5)
A14	12:04:50.375	SEC / 4672	83548	Special privileges / SYSTEM
A15	12:05:06.591	SMN / 1	28365	HOSTNAME.EXE
A16	12:05:06.650	SMN / 1	28366	HOSTNAME.EXE
A17	12:05:06.715	SMN / 1	28367	HOSTNAME.EXE
A18	12:05:06.747	SMN / 1	28368	whoami.exe
A19	12:05:07.471	SMN / 1	28369	HOSTNAME.EXE
A20	12:05:07.500	SMN / 1	28370	whoami.exe
A21	12:05:07.886	SMN / 1	28371	cmd.exe
A22	12:05:07.953	SMN / 1	28372	reg.exe
A23	12:05:07.993	SMN / 13	28373	Run key SetValue

Cross-feed matches: **L1 = A21** (Sysmon record 28371, Wazuh rule 92052); **L2 = A22** (28372, rule 92041); **L3 = A23** (28373, rule 92302). Thus three alert objects do not represent three additional endpoint actions.

Input integrity references (SHA-256 of the supplied JSONL files):

```
Scenario_3_Archives.json   6b7281cb3d313b4b95c94560880c89826789bdb6d6a35e2ffa83576efbcb64a1
Scenario_3_Alerts.json     f49ade4787d428eceafb0b942d032c0a2e109fdd7467ceeab81011d7b3a65368
```

5. Timeline of Events

Chronological event times below are UTC on **27 October 2026**. Sysmon process and registry rows use **eventdata.utcTime** (millisecond precision); other rows use Windows **systemTime**. The Wazuh collection timestamp is later than event occurrence.

UTC time	Group	Event	Ref
12:02:05.825-12:02:06.024	Background	Security service-logon/privilege events and six Credential Manager reads (client PID 7472); no established relationship to later process chain.	A1-A8
12:02:53.350	Background	SYSTEM MoUsoCoreWorker.exe started by UsoSvc-related svchost.exe.	A9

UTC time	Group	Event	Ref
12:03:53.553	Background	Kernel-Power 566 session unlock transition.	A10
12:04:50.329-12:04:50.376	Background	Input-HID transition; SYSTEM logon/privilege events; CameraMonitor svchost.exe launched.	A11-A14
12:05:06.591	Related	PowerShell parent (PID 12276) creates hostname.exe PID 2772.	A15
12:05:06.650	Related	Second hostname.exe, PID 11160.	A16
12:05:06.715	Related	Third hostname.exe, PID 1148.	A17
12:05:06.747	Related	First whoami.exe, PID 8076.	A18
12:05:07.471	Related	Fourth hostname.exe, PID 8600.	A19
12:05:07.500	Related	Second whoami.exe, PID 8176.	A20
12:05:07.886	Related	cmd.exe PID 7912 launched by PowerShell with cmd /c REG ADD of HKCU\...\Run.	A21 / L1
12:05:07.953	Related	reg.exe PID 2824 created by cmd.exe PID 7912; same requested Run-key write.	A22 / L2
12:05:07.993	Related	Sysmon 13 confirms SetValue of Atomic Red Team to C:\Path\AtomicRedTeam.exe by reg.exe PID 2824.	A23 / L3

Timing note: the cmd.exe start to confirmed registry SetValue spans roughly **107 ms**; the reg.exe start to SetValue spans roughly **40 ms**. Those intervals and explicit parent-GUID links support the single command-chain interpretation.

6. MITRE ATT&CK Mapping

Mappings below are behavior-based interpretations of the supplied events rather than adoption of Wazuh labels. They identify what activity is supported; they do not imply the actor's identity or malicious intent.

Tactic	Technique	Evidence and bounded rationale
Discovery	**T1082 - System Information Discovery**	hostname.exe invoked four times under the common PowerShell parent [A15-A17, A19]. Consistent with querying the computer name; output not captured.
Discovery	**T1033 - System Owner/User Discovery**	whoami.exe invoked twice [A18, A20]. Consistent with discovering execution identity; output not captured.
Execution	**T1059.003 - Windows Command Shell**	Explicit cmd.exe /c command issued to add a Run-key value [A21 / L1].
Persistence	**T1547.001 - Registry Run Keys / Startup Folder**	Per-user CurrentVersion\Run value Atomic Red Team set to executable path [A23 / L3]. Actual later autorun unobserved.
Defense Evasion (technique only; tactic use not established)	**T1112 - Modify Registry**	reg.exe plus Sysmon SetValue show direct registry modification [A22-A23]. The modification is proven, but an evasion purpose is not proven.

Mappings not supported: Wazuh rule 92041 lists **T1027 - Obfuscated Files or Information**, but the observed Run-key data is cleartext and no encoded content is evident. The privilege-escalation tactic listed with rule 92302 is also not proven by a high-integrity process or unrelated SYSTEM logon records. Although powershell.exe is a demonstrated parent, no PowerShell invocation arguments or script body are present to characterize additional PowerShell-specific behavior independently.

7. Overall Investigation Conclusion

Observed facts: One Windows user context launched six hostname/whoami discovery commands from PowerShell, then started cmd.exe and reg.exe; Sysmon independently observed that reg.exe set a user Run-key value named Atomic Red Team to C:\Path\AtomicRedTeam.exe. Exact process GUIDs, PIDs, usernames, logon ID and tightly clustered timestamps tie these records together [A15-A23, L1-L3]. The three alerts are duplicates of already archived underlying Sysmon activity.

Inference and significance: The Run-key write is consistent with preparing logon-based persistence. Its practical effectiveness and any hostile objective remain unresolved: the dataset contains no evidence of the target executable on disk, its execution, a subsequent user sign-in, lateral movement, external communications or damage. Hostname/whoami behavior preceding the write makes a single local activity sequence more likely, but does not by itself prove malicious reconnaissance. The lab framing and test-like value name are context, not a basis for presuming authorization.

Overall confidence: HIGH in the reconstructed local sequence and confirmed registry modification, because Sysmon process creation and registry-change records corroborate one another with precise GUID and command-line matches. Confidence in the narrower inference of **persistence intent is medium-to-high**; confidence that the entry achieved working persistence or that an unauthorized compromise occurred is **low** on these records alone.

Investigation boundary: this report evaluates the provided telemetry only and does not assert that unlogged actions were absent.